



Understand Future Forces & Generate New Value

2024

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“The Future of Packaging is Now.”

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Are We Prepared?



For People, Planet & Profit

The relationship between packaging and several product categories is a nuanced and symbiotic one. This translates in to a simple fact that innovation in packaging has far reaching consequences on business ecosystems at large. While the impact of growth in the packaging space is evident, what is often overlooked is the importance of recognizing challenges that are likely to come in the way of unlocking opportunities for innovation.

To dive deeper into this, Packfora, a global leader in packaging solutions, hosted Packforum 2024. During the event, we explored future trends and the key preparations needed to seize the market's full potential.

The insights shared at the event have been compiled in this whitepaper, reinforcing our commitment to establishing a thought leadership platform in the industry.



We share with you a view of innovation in packaging with five key points to keep in mind:

Balance

We are faced with a commitment to sustainability and the structure of regulations, but at the same time, customer engagement is paramount. Our goal is to strike a balance between what we are expected to do, what we are able to do, and what we need to do.

Strategy

Packaging today has several perspectives to incorporate into design and structure – from functionality to forecasting future trends to understanding what today's consumer wants. To reach our audience, we need strategic packaging solutions that take in to account several factors.

Change

Digital advancements are causing paradigm shifts across the board – the packaging industry is no different. It has become critical to be able to stay

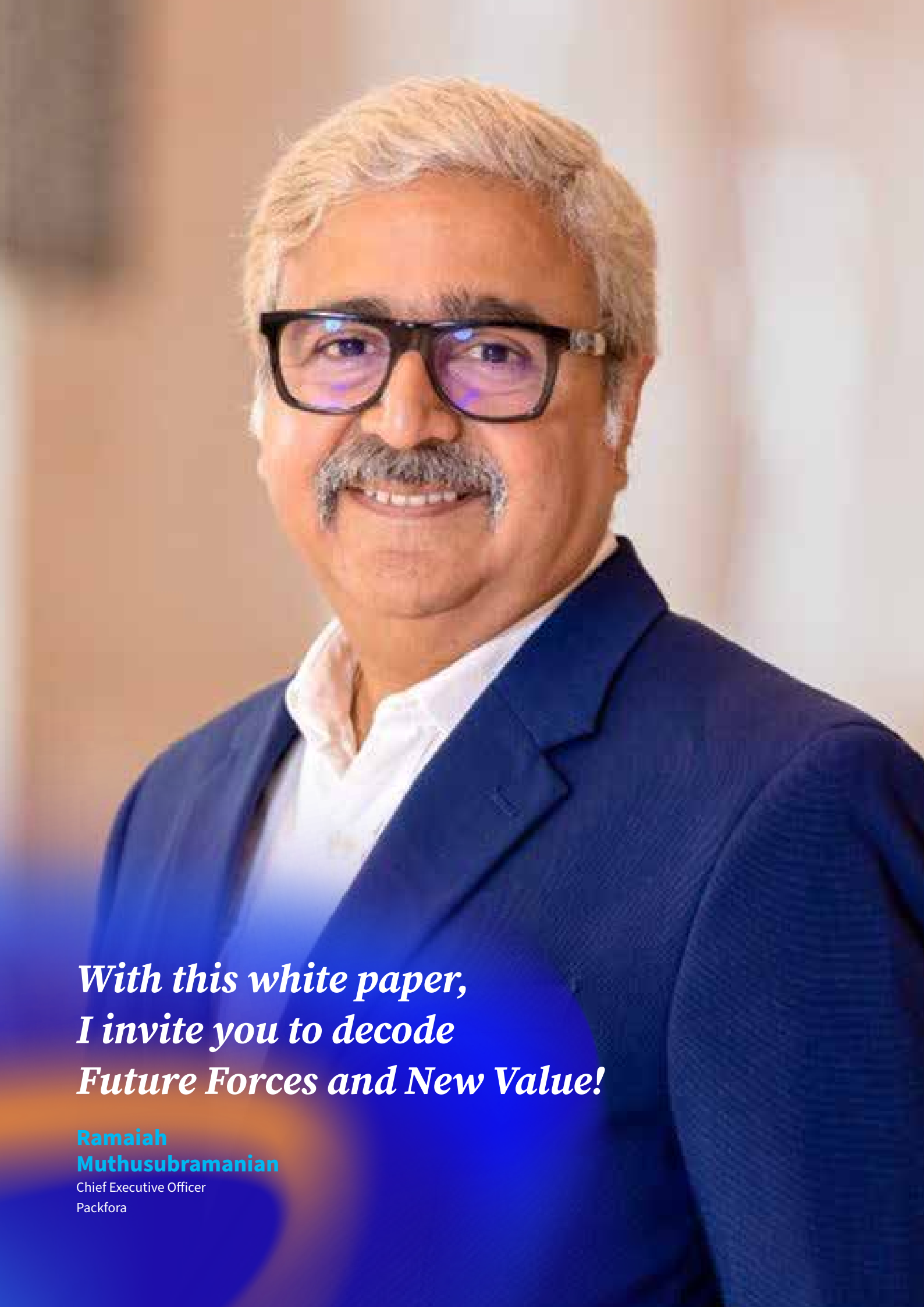
ahead of the curve to create solutions that help companies brace themselves for tectonic shifts in the market.

Purpose

The role of design in packaging is well known, but less acknowledged is its functionality. The next wave of growth brings in an entirely new host of products with their own unique demands in terms of functionality. We must ensure that the innovative solutions we create cater to supporting the product's care.

Value

Packaging solutions are gaining visibility now, more than ever before, particularly because social media reaches millions of people with a single click. This has made it important to ensure that how a product is presented contributes to its overall value proposition.



*With this white paper,
I invite you to decode
Future Forces and New Value!*

**Ramaiah
Muthusubramanian**
Chief Executive Officer
Packfora



A Word from the CEO

There has never been a more challenging time than now for the global packaging industry. And concurrently, there has never been a more exciting time for us. We're seeing a transformation in what the consumer seeks, what organizations seek, and what the environment seeks. Being agile and innovative in the face of these needs is no longer a "good to have"; it is a "must have."

Interestingly, the changes needed in packaging solutions are not limited to one company alone or a group of companies; what we face are 'mega problems'. Collaboration, to circumvent these obstacles, is key. For example, packaging that doesn't use traditional plastic or packaging that is relevant to changing economic factors, these problems require collective action. And that's why we, at Packfora, created Packforum.

Our goal was a simple one: Let's find opportunities in adversity and solve bigger problems via joining forces. The response we received to doing this was overwhelming. Across the board, we found a desire to see the industry become more connected. The binding factor is wanting to use packaging to create a competitive edge, to make packaging central to a brand and not simply a carrier.

From the event, I found there to be several interesting revelations:

➤ Sectors across the board – from FMCG to Pharma to consumer electronics everyone is getting impacted by digital changes – this is a whitespace for everyone, and we're all actively building innovative solutions that incorporate new age materials, future-fit tech features and more.

➤ The consumer cares about the environment. 5 years ago, this differed significantly from person to person but over the last few years, eco-consciousness has become pervasive. Eco-friendliness now prominently features in consumer choices.

➤ The risks associated with companies launching new and disruptive solutions has become higher the visibility brands receive is widespread, putting everything we do under the microscope, even more so than before. It has made the opportunity for experimentation that we must approach innovatively, but also carefully.

➤ The industry wants to connect as we all realize that polarizing consumer demand, environmental regulations, digital growth, and supply chain issues - all of these are challenges that we can ably tackle through relevant discussions, forums, and events.

As I look ahead at what packaging is going to evolve into years from now, I feel deeply enthused. If we are able to understand how to turn obstacles in to opportunities, together, there is so much scope for innovation, especially with new-age tech such as AI and ML on our side.

Packforum is here to stay, and next year will be focused on bigger and bolder themes with experts across the value chain – beyond brand owners. The exchange of ideas, collaboration, and engagement that became a hallmark of the event has resulted in valuable insights that aptly capture what we can expect next, how we should prepare for it, and why changes in consumer choices can be used to our advantage.



What's Ahead

The pages ahead represent thoughtful strategic insights developed in collaboration with Packfora's partners. Together, we have co-created a central theme addressing the key trends.

Smart commerce is built on digital as the foundation. It is rapidly redefining routes to consumer reach and experience.

Smart Commerce



Regulation



The world is moving from banning to the polluter pays. Therefore, business continuity is hinging upon preparedness for executing mandatory or enforced changes.

Next Generation Recycling & Materials

Creating a waste-free world, with new perspectives & technologies.



End-to-End, Digital/AI based Life Cycle management is a necessity for consumer experience, consumer and planet safety.

Smart Supply Chain



Blurred Boundaries

Unlock big benefits by reimagining extensions of proven packaging formats in new categories.

Global Factors

Polarized economy = Extreme consumer preferences (Shrinkflation to hyper-personalization).



Refill Revolution

Refill systems: a potential triple bottom line disruptor. Waiting for the right consumer hook to scale up.



Inclusivity and Diversity

Inclusive Design for everyone is the new Mantra. More inclusive design = More business growth.



Our constant endeavor to stay ahead of trends has revealed that the world is moving into two pivotal areas: Future Forces and New Value. These areas represent the critical shifts we believe will drive innovation and growth in the packaging sector.

With this whitepaper, our aim is to deliver in-depth insights by identifying key themes and challenges, focusing on the most pressing issues and emerging trends that demand attention. Our goal is not merely to inform, but to inspire action and guide strategic decisions that empower businesses to navigate change and seize new opportunities in an evolving landscape.

Future Forces

From a pandemic to a war, if history has taught us one thing, it is that changes in circumstances come with little warning. The question then that begs to be asked is, “How do we minimize business impact?” For packaging, this means understanding what future forces could be and ascertaining ways to prepare for them.

Future forces are not limited by negative events; in fact, the trick lies in how to interpret them. Reviewing potential changes with an eye to identify opportunities for innovation within them will allow us to not

only sustain growth but amplify it.

Consider each future force as a checkpoint encountered by every company in the packaging space. At the checkpoint’s end, will be two options – first, a path of least resistance, resulting in status quo, and second, a path with difficult, strategic transformations, resulting in a new identity for the organization. It’s important that we choose the latter, because it is doing that will prepare us to find unique ways to brace the forces of the future.

What does this mean for the packaging industry?

Future forces in the packaging industry are spurred by the dynamic nature of regulations and consumer demands. The effect of the world changing is evident on both, governments and the consumer.

From laws that ban the use of certain materials, to the introduction of AI in to the supply chain, to consumers choosing to shop with clicks of their mouse versus steps in a shop, we’re seeing consumption patterns and preferences evolving. It is critical for packing solutions to follow suit.

Future Forces

Regulations

We're seeing the repercussions of excessive plastic consumption everywhere around us. The more obvious impact is by way of its presence in our seas, in our soil, and in our groundwater. What is often overlooked is the less apparent ways in which it is hurting us.

For example, in Ghana, in 2015, when plastic blocked the capital city's drainage, it resulted in complete flooding of Accra's central business district. The damage caused extended far beyond property: stagnant pools of polluted water spurred hundreds of cases of malaria and cholera. The fact that Ghana is not alone has made the Global Plastics Treaty amongst the most pivotal we've ever seen. The world, including the business community at large, is looking towards governments to understand how specifics of the treaty as going to come in to play. The proposed pollution tax will reduce plastic products available and promote recycling. How we create product packaging is likely to go through a paradigm shift and therefore, new costs.

To optimally insulate ourselves against rising costs, the goal should be adapting to regulatory changes that will accompany changes in the environment. One of most efficient ways to adapt to changes is to anticipate them and, in this case, it means closely following global dialogues on the topic.

Minderoo Research estimates that a US\$60-90 per tonne fee on primary polymers is needed to cover the US\$350-500 billion gap that will enable emerging economies to effectively adopt changes related to the adoption of recycled polymers. Virgin plastic is cheaper, compared to recycled plastic and to ensure that recycling measures adopted across the world, monetary support is critical for emerging economies.



2.1

2.1: Recyclable materials replacing plastic waste can have a massive impact | 2.2: Plastic cutlery poses a major challenge

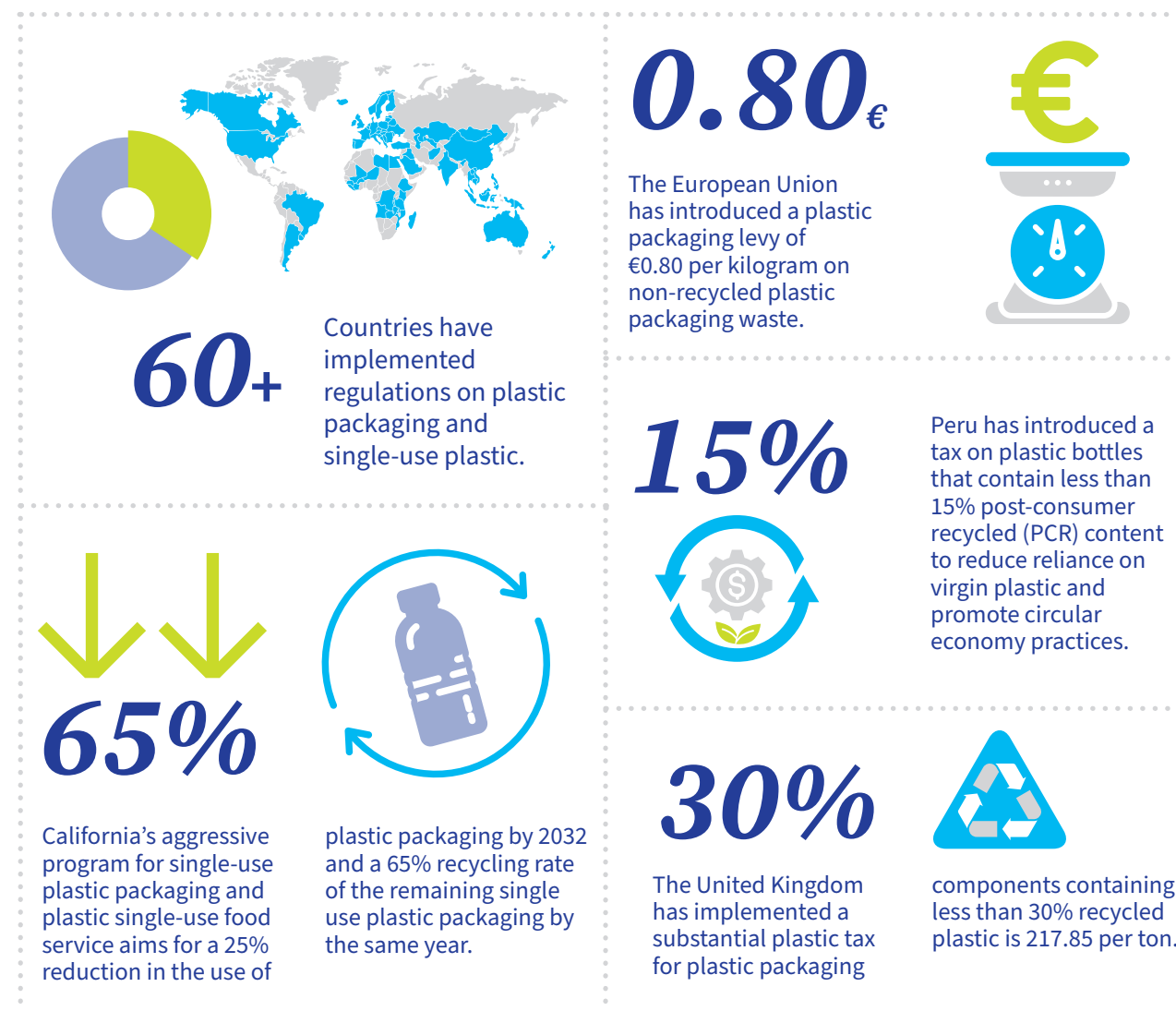


2.2



2.3

Plastic Tax



Changes in regulations do not affect businesses uniformly, while some could require sweeping overhauls of production, others could accommodate new rules with just a few tweaks. For example, in the CPG industry, using recycled content is becoming increasingly mandatory and feasible due to less restrictive safety regulations. In contrast, the Pharma industry has limitations on PCR adoption due to stringent safety and contamination controls.

Similarly, CPGs are often required to report on the amount of packaging used, recycling rates, and progress towards waste reduction targets, whereas Pharma companies may need to report on the effectiveness of take-back programs, volumes of pharmaceutical waste collected, and disposal methods. Assessing changes in light of specific business operations is critical.

The Bottom Line

Whether changes in regulations are enforced or recommended will matter, but at the same time, we are seeing a shift in sentiment wherein there is a clear preference developing for sustainable packaging that aligns with regulations. To gain a competitive edge, packaging solutions not only need to become greener but also need to be available quickly.

Future Forces

Next Generation Recycling and Materials

There is one, and only one way to develop packaging solutions that align with dynamic changes in consumer preferences and government regulations: new recycling technologies and new materials. The focus on creating a waste-free world is inevitable and to enable this we need fresh perspectives bolstered by innovation.



3.1: Investment in Next-Gen Materials Rebounded to \$500 Million in 2023

Are seaweed polysaccharides the next answer to single use plastic packaging?



Can Bamboo injection molded packaging revolutionize packaging?



What do new non - polymer performance coatings mean for paper?

Next Generation Recycling and Materials

Advanced recycling methodologies must account for not only enhancing sustainability but also improving efficiency. While approaches such as chemical recycling to solvent-based recycling can be incorporated into packing solutions – the challenges involved can be significant.

New technology is not limited to recycling methods alone. These will also be responsible for the creation of the next generation of packaging materials and design. For example, in the case of the Food & Beverage industry, we are beginning to see the adoption of ‘Smart Packaging’ solutions. These detect indicators such as food freshness and temperature history. Being cognizant of these factors will reduce wastage of perishable products. Packaging brand, Notpla’s solutions reflects this ethos. The brand has introduced a revolutionary Seaweed-based packaging for pods that is entirely edible. This means that Notpla’s gel pods are biodegradable at multiple levels: i) if they are consumed through eating, they do not pose any environmental risk ii) if not eaten, they will

naturally break down in the environment within weeks and iii) if disposed in the sea, instead of toxic waste, They act as food for other living organisms.

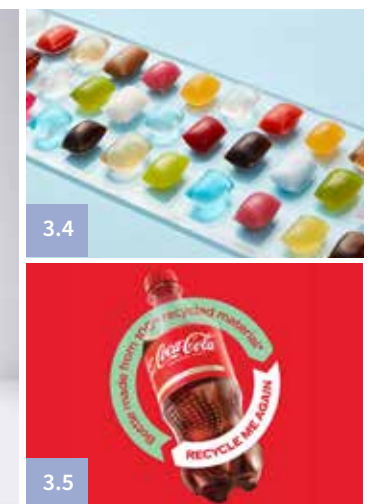
Renewable materials are an indispensable part of new-age packaging. Bio-plastics can now be derived from renewable sources like plant starch, cellulose, vegetable fats, natural sugars, and they can also be made from marine algae, which is a renewable natural resource that absorbs carbon from the atmosphere, have a lower carbon footprint, particularly when pegged against traditional petroleum-based plastics. Bio-PET (Bio-based Polyethylene Terephthalate) is a bioplastic partially derived from renewable resources, typically sugarcane or corn, offering a sustainable alternative to conventional PET. It retains the same properties and recyclability as petroleum-based PET, including durability, transparency, and resistance to moisture. Although only about 30% of its composition comes from renewable sources, ongoing innovations aim to increase this proportion, enhancing its environmental benefits.



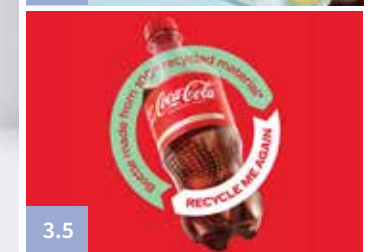
3.2



3.3



3.4



3.5

3.2: Edible packaging, made from materials like seaweed and starch, offers an eco-friendly alternative to plastic | 3.3: Coca-Cola has begun putting 'label-less' Sprite bottles on shelves to reduce waste | 3.4: Notpla's Ooho packaging aims to reduce plastic pollution while providing convenience at events like marathons | 3.5: In New Zealand, all plastic Coca-Cola bottles smaller than 1 litre will be made from entirely recycled materials



The future of new recycling technology in packaging and technology is overwhelming by way of its scope. However, when strategically approached, this very scope allows for endless possibilities, across industries. Packaging companies must closely examine related nuances and create solutions that match multiple criteria: what they are able to create cost efficiently, what suits clients' product categories and whether proposed solutions can be sustained in the long run.

Future Forces

Smart Commerce

The journey of a consumer, from entering a store to making a choice, is a process that most packaging companies are well-versed in, particularly how consumers respond to packaging on a shelf. However, the way consumers make choices is evolving. Today, Gen Z is more likely to discover CPG products online rather than in a store. How a product is perceived will differ, and that perception relies heavily on packaging.

The statistics are staggering – the number of gadgets in use has exceeded 25 billion and we are expected to see a 75% rise in e-commerce. With more clicks being made to buy, the more companies realize that packaging has to change its appeal so as to make an impression across a screen. Even more important is the fact that social media is becoming the source of product discovery. How products appear on TikTok, Instagram and other varies from the manner in which they are designed to stand out on a shelf.

Improving the immersiveness of the product experience has been made possible by the fact

that smart technology is now in the hands of most consumers. Alcohol brand Jack Daniel's capitalized on this by launching an Augmented Reality (AR) app called the Jack Daniel's AR Experience. They achieved their goal of bringing whiskey bottles to life through mobile devices. Scanning a code on the bottle launches users in to a brand story including an overview of the whiskey's distillation and aging process, and stories about the brand's founder. In doing this, Jack Daniel's was stretched past limitations of traditional advertising and optimized the real estate on the bottle's packaging label.

25 Billion

gadgets in use globally

Gen-Z

is more likely to discover CPG products on search than on store shelves

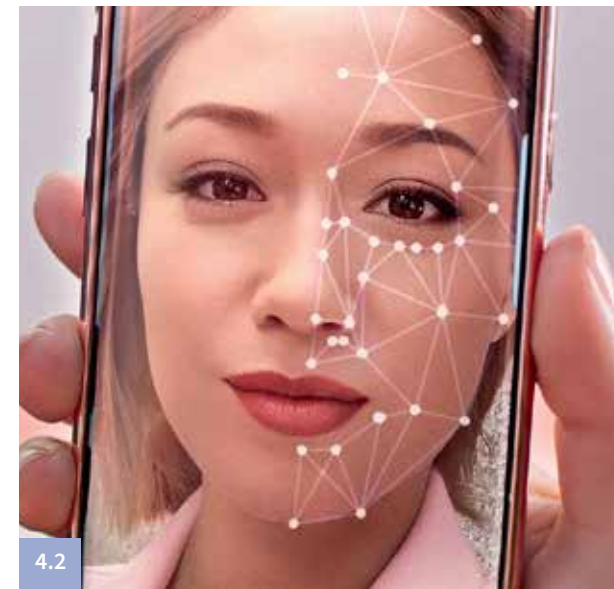
75%

expected rise in E-commerce



4.1

4.1: Jack Daniels has introduced an AI experience for its product



4.2

4.2: L'Oréal made available its AR-powered packaging

Developing packaging that is suited to smart commerce focuses on a number of factors, spread across metrics that are we are both, accustomed to such as visually appealing design and entirely new ones such as the “unboxing experience”. Speaking to the latter, packaging that provides a noteworthy unboxing experience is likely to positively influence social media sharing. Similarly, introducing the concept of personalization is possible through smart commerce. Creating packaging that can lend itself easily to hosting the buyer's name or a special message once purchased online adds a lot of value to the consumer thereby enticing him/her to shop online.



4.3

4.3: Deliveroo offers customers a new shopping experience with the option to order via digital kiosks, the Deliveroo app, or for rapid delivery, expanding its commitment to convenience and sustainability through unique partnerships | 4.4: Coca-Cola's limited-edition "Happy Tears" drink marks the company's first use of the Tik Tok platform for direct sales

Smart commerce is built on digital as the foundation. It is rapidly redefining routes to consumer reach and experience.

A terrific example of how synergies can be created to drive significant value through packaging is when L'Oréal, arguably one of the most well-known cosmetics brands, joined hands with Amazon, the world's largest e-commerce platform. Through the collaboration, L'Oréal made available its AR-powered packaging on Amazon. This gave users a chance to scan makeup products to activate a deeply immersive virtual try-on experience leaning on the brand's ModiFace technology. Heightened interactivity increases consumer confidence in purchasing beauty products online.



4.4

The Bottom Line

Smart commerce must be viewed as a pool, albeit a new and relatively uncharted one, of endless opportunities for innovation in packaging solutions. The scope for productivity as a function of people shopping online is exciting, once it is understood and evaluated holistically.

Future Forces

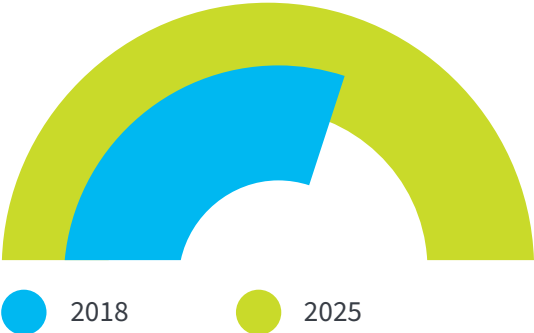
Smart Supply Chain

Traditional supply chains, as we know them will cease to exist within just a few years. New-age tech, from AI to machine learning, is making supply chains more transparent, efficient and accountable. The packaging industry stands to benefit from this spike as we can provide access to tech-driven design elements that map directly to smoother supply chain management.



5.1

5.1: Barcode Scanning in Supply Chain Operations



10.1 Billion

Research conducted by ReportLinker found that the AI in supply chain market is expected to reach \$10.1 billion by 2025, up from \$730.6 million in 2018, at a CAGR of 45.55%.

The most important advantage that packaging can provide supply chain management is time-saving. Smart packaging can generate real-time information about inventory levels, allowing retailers to set notifications that would spur product restocking. This, in turn will help companies completely do away with manual verification and aid in the automation process. The fact is that profit margins for most sectors, particularly the retail industry, are extremely tight. Supply-chain friendly packaging that stands to limit disruptions is an investment most organizations are willing to make, keeping in mind long-term benefits.

According to a report by the Capgemini Research Institute, 66% of customers choose to purchase products or services based on their “environmental friendliness.” Companies can claim to be more environmentally friendly if they are able to ensure less product wastage.



5.2: Henkel pioneered digital watermark technology with its Vernel product range | 5.3 Nestle’s Mousline relies on QR code technology to give consumers a clear insight in to its green practices

Matching product demand and supply is a difficult task, but incorporating technology, particularly AI, can make this much easier. Receiving and delivering items on time, replenishing them in a timely manner on shelves, monitoring delivery through QR codes all of this can be achieved through smart packaging.

Nestle, with their product Mousline, a mashed potato brand, was able to provide consumers a very clear insight in to green practices. By scanning a QR code on the Mousline packaging, customers could access detailed information about the product's entire supply chain. This included data such as the variety of potatoes used, production dates, storage times, quality control measures, and the exact location of warehouses where the product was stored. Consumers could also learn about the farmers involved in producing the potatoes, highlighting the authenticity and traceability of the product.

The Bottom Line

The key to minimizing errors in supply chain management, and thereby reducing wastage, is to limit manual intervention as much as possible. This is why there is a constant effort across industries to achieve this goal. Packaging design can help expedite this process for many companies. By embracing the power of new-age technology, we can make significant strides in this endeavour

New Value

Changing winds are ahead of us, and these are coming in the form of shifting business and consumer needs. The way to gain a competitive advantage in the face of fluid market dynamics is identifying opportunities for new value creation. Offering new value to the consumer or business, in the form of conscious packaging that directly maps to evolving preferences can be challenging, but it is of paramount importance. Companies see traditional packaging solutions as not only obsolete but also a factor that contributes directly to growth. A revamped

approach to how they present a product is being viewed as an opportunity for growth and at the same time, older solutions that no longer relate to modern requirements are being considered a potential hindrance to success. This is fast-making the adoption of new value propositions part of risk mitigation strategy. Generating new value can take the form of proactively exploring brand equity, broadening target markets, and making most of an increasingly borderless business ecosystem. These avenues are all opportunities to make conscious choices.

What does this mean for the packaging industry?

The power of innovation is indispensable to our quest to turn challenges into opportunities for new value creation. By taking unconventional approaches, fueled by creativity, packaging solutions can respond to modern sensibilities. We can create packaging solutions that give products not only the ability to cope with changes but also thrive in a dynamic environment.

New value gateways focus on leveraging important tools that are at our disposal, such as AI, to fundamentally transform not only the visual appeal of packaging but also the effect it has on top and bottom-line growth. Through identifying and optimizing areas for value creation, the impact of packaging can be made clear.

New Value

Blurred Boundaries

There are big benefits waiting to be had by reimagining extensions of proven packaging formats in existing categories. We must embrace the concept of “re-learning” in order to bring a fresh perspective to the tried and tested. There is a sense of security that comes with taking a cookie cutter approach, but the speed at which preferences and needs are changing makes agile innovation an indispensable part of operations.

The role of packaging has evolved to the point of it now being a ‘brand communication vehicle’. As a result of this, packaging can be used for much more than simply storage: it can add to the brand’s story and messaging. A fitting reflection of how some organizations have managed to creatively think about new ways to adapt their product presentation to enhance their brand’s perception is the packaging of ‘Method’.

Method is a household and personal care cleaning brand. They are known in particular for having plant-based, environmentally friendly products. Leveraging this brand perception, Method adopted sleek, reusable bottles for eco-friendly cleaning supplies that boosted their green cleaning market share by 15% showing how innovative packaging can lead to growth in emerging segments. This commitment to sustainability resonated with environmentally conscious consumers and aligned with the growing demand for eco-friendly products. Similarly, Graza, a modern olive oil brand, disrupted the market which traditionally uses glass bottles. They introduced recyclable plastic squeeze bottles which have lower transportation emissions – this measure further enhanced their recall as a sustainable brand.

Market differentiation is another key metric that can be achieved through understanding how products cross category boundaries. Liquid Death, a brand of water, did exactly this by creating packaging for its cans that resembles that of energy drinks. In doing this, the brand was able to

stand out in a very crowded bottled market and increase the gravitation of the consumer towards its product.

At the same time, it’s relevant to note that packaging doesn’t have to be overhauled entirely. Existing packaging can be revisited through the lens of how it can be modified to reflect packaging relevant to an entirely new product category. The re-purposing of designs from other categories reduces development time while providing fresh solutions to meet changing consumer needs. The role of innovation in achieving speed to market will play a big role here.

15% *Method Products’, sleek, reusable bottles for ecofriendly cleaning supplies boosted their green cleaning market share by 15%.*

50% *Seventh Generation's milk-carton like packaging for its cleaning products cut plastic use by 50% and boosted customer loyalty.*



Speed To Market.
Shelf standout in a competitive space.

Blurred Boundaries



6.1 & 6.2: J&J and Lactel launched a refillable product range offering affordable, eco-friendly solutions in recyclable 1L carton, combining convenience with a familiar, user-friendly format | 6.3 & 6.4: Liquid Death packaged their water in to cans resembling energy drinks | 6.5: Cleancult products are biodegradable, zero waste, green cleaners 6.6: Seventh Generation's Zero Plastic Homecare line eliminates plastic packaging by using dry products like powder hand soap and dishwasher tablets in recyclable steel containers

The Bottom Line

By looking beyond what we know, what has been used and what has worked in the past, we can create packaging solutions that combine different aspects such as product functionality, category, and more. Success in doing so will help us make packaging a major point of differentiation for consumers.

New Value

Inclusivity and Diversity

It is simple: for a brand to grow, more people need to use its products, putting inclusivity and diversity at the heart of strategic growth plans. In the case of packaging, this means design and functionality which is relevant and appealing to a larger audience, spread across cultures, income strata and other typically differentiating categories.

In 2022, the Design Age Institute estimated that “Three in five people in the UK over 55 find opening packaging to be a big stress factor.” It would then be safe to assume that at least 1 of these 5 people are likely to not re-use a product that had been responsible for a negative experience. Easy-to-open and close products are fast becoming a requirement for brands as consumers tend to be drawn towards product that can prominently showcase that feature of their packaging. Senior citizens, for instance, would lean towards packaging that is straightforward for them to manage.

Convenience is another factor in packaging that contributes to how inclusive it is. Brands in the oral care for example are making it priority to use

tubes that dispense the right amount – the packaging herein would either entice or incentivize the consumer. Using a daily product that makes life more convenient is likely to draw in more people, across consumer demographics, and help the product grow its market share.



Three in five people in the UK over 55 find opening packaging to be a big stress factor.



7.1: Investment in Next-Gen Materials Rebounded to \$500 Million in 2023



7.2



7.3



7.4

7.2: Kellogg Europe added world-first technology to its cereal boxes to make them accessible to blind and partially-sighted people | 7.3: Degree is developing a set of customizable accessories designed to help individuals with upper limb disabilities more easily use deodorant | 7.4: P&G's Olay face cream's design features an easy-open winged cap, extra grip raised lid and a high-contrast product label and Braille text that reads 'face cream'

Packaging today has to be need-sensitive, especially as technology has become a major enabler allowing us to provide support where it is most needed. For instance, key segments of audiences are visually and/or physically impaired in some way. Various brands create packaging that is bold enough to capture the interest of these audiences or adopts extremely innovative packaging solutions such as sensory enhanced bottles. Ensuring design that is suited to these consumers speaks to a brand's desire to be inclusive and diverse, which adds a lot of value from both, perceptual and commercial perspectives.

Degree, a leading antiperspirant brand, introduced precedent-setting packaging that appeals to and helps a significantly larger audience than the average product in this category. 'Degree Inclusive' became the first “adaptive deodorant” designed for accessibility. It features a hooked container for one-handed use, improved grip, magnetic closures, and a larger roll-on applicator, with instructions in Braille. This innovative packaging style represents a significant advancement for the industry, revolutionizing the consumer experience.

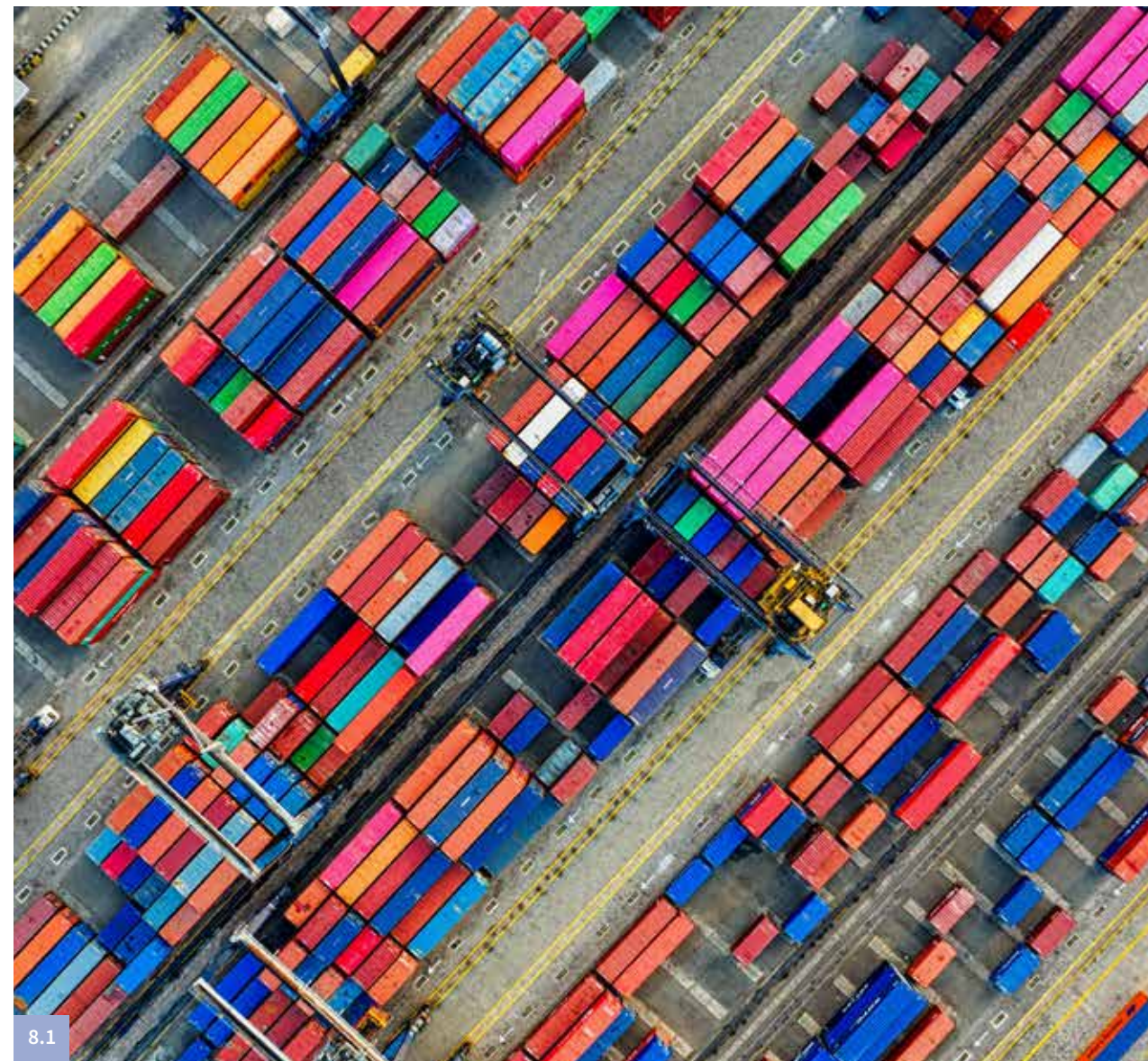
The Bottom Line

Inclusivity and Diversity have become fundamental cornerstones of innovative packaging design, prompting organizations to implement social change in truly unique ways.

New Value

Global Factors

Balancing factors of the global economy can be challenging for businesses. With inflation on the rise, we're seeing a very obvious and expected demand for lower prices. To respond to this, 'shrinkflation' has come in to play, wherein many organizations have either kept prices the same or reduced prices, but reduced the quantity of the product provided. Making this shift is difficult as it calls for an overhaul of packaging, sometimes at different intervals based on market response. An important frontier for new value becomes the ability to provide organizations with packaging solutions that are agile and adaptable.



8.1

8.1 Barcode Scanning in Supply Chain Operations

In the United States, some major brands reduced product sizes by over 20% in 2023 without reducing prices; “shrinkflation” averaged 11.2% among selected national grocery brands. A fitting example of this is popular potato chip brand, Pringles. We saw Pringles subtly introduce new smaller sizes of packaging without changing the price point. The packaging of the products did not change, despite the container size being different for packaging to provide organizations flexibility of change is critical.

To battle inflation, the introduction of ‘limited edition’ products can be particularly helpful. Many brands have found that bringing to the market a new product, unseen before and available for a set period of time has a deeply positive impact on consumer interest. Moreover, the result perceptual exclusivity of the product allows for a higher price point. This is an opportunity for incredible innovation in packaging solutions.

Design for limited products must reflect a number of factors, ranging from use of new-age materials, incorporation of technology and a look and feel suited to the nature of the product. Coca cola has achieved great success with this model, wherein there are several flavors that have become so popular as limited editions that they are now included in their regular product line.

Packaging that improves durability of products can also play a key role in helping companies cope with tighter purse-strings. Amcor, global packaging solutions company, helped Popcorn Shed improve customer retention through packaging that was not only far more eco-friendly but also extended the shelf-life of their product. With Amcor's new solution, the brand's popcorn stays fresh for as long as 12 months. A value proposition of this nature naturally attracts the consumer as it means investing in a product that can be enjoyed for a longer period of time.



8.2

8.2: Pringles changed their packaging without compromising on price point | 8.3: Coca Cola's limited edition products have seen great success with many finding a permanent place in their offerings 2023



8.3

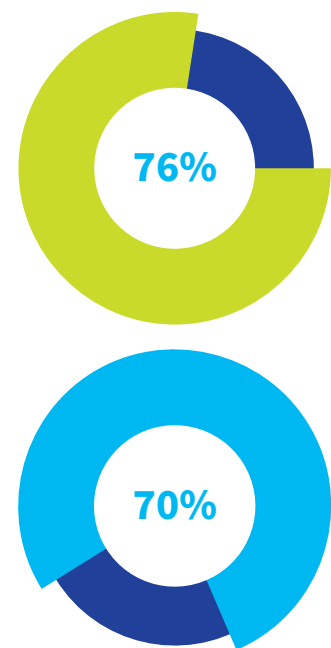
The Bottom Line

The need of the hour is to have technologies enabling companies to thrive in complexity as consumers are looking for frugal life hacks to maximize what they can get from their budgets.

New Value

Refill Revolution

Sustainability is important to a happy and healthy future – this is an indisputable fact. What is lesser known is how much this truth factors in to consumer decisions today. What was earlier a ‘nice to have’ is fast becoming a ‘must have’ based on how much consumers are leaning towards greener packaging solutions. As a function of this, packaging the 21st century will therefore take many forms.



76% of children in the US & 70% in UK feel it's important to them and their lifestyle to become more environmentally friendly.

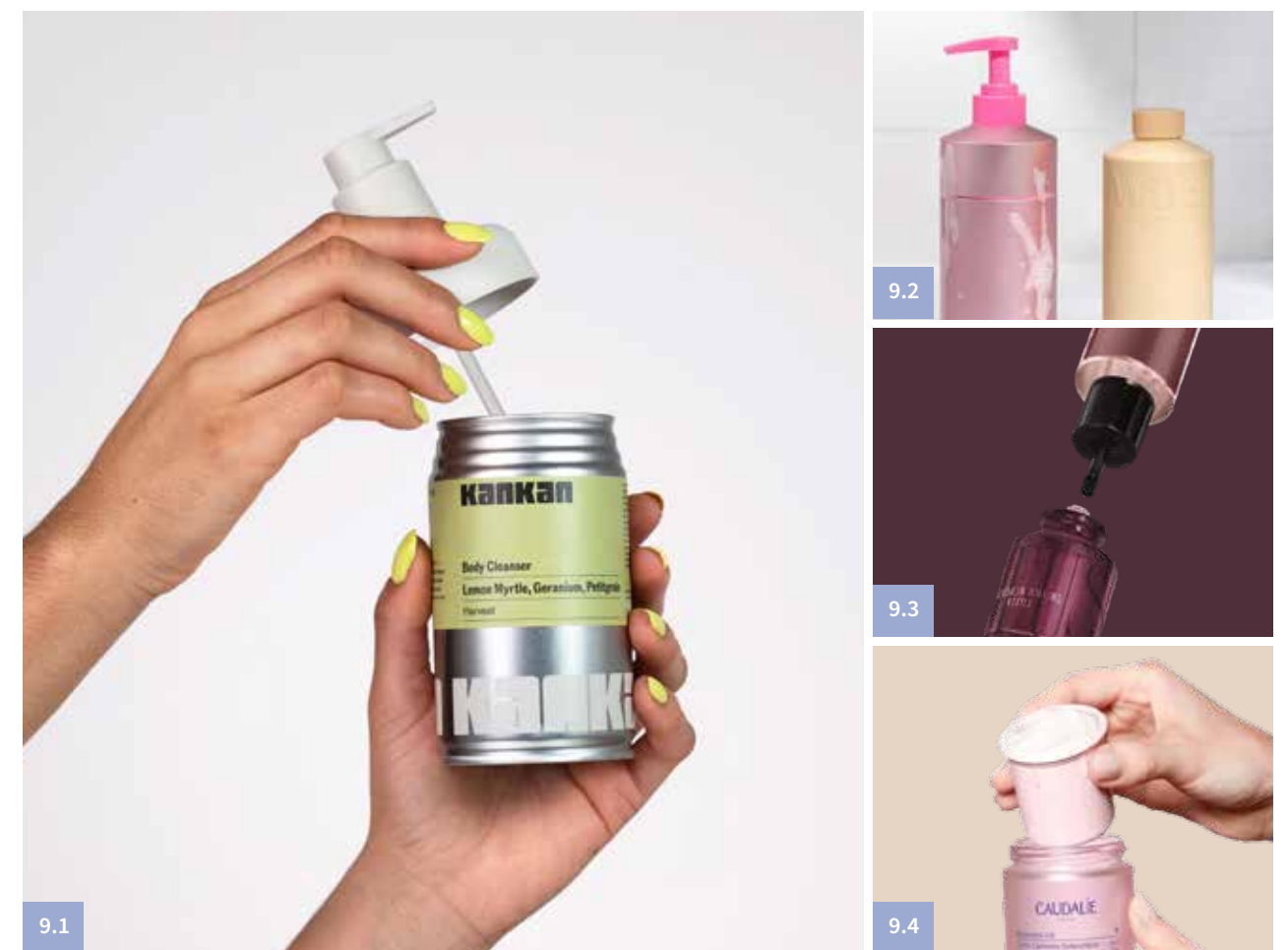
Predominantly, change in packaging can be offered by way of containers that enable multiple use by way of refills. The adoption of refillable containers has become prevalent across sectors, including beauty and makeup which has been long considered amongst the most wasteful industry in terms of packaging. The added advantage is that product refills cost less too making the first-time purchase of the reusable container all the more enticing. A 2024 trend report, “*The Future Unboxed*” explained how cleaning brand Puracy introduced the

CANcel refill system, which includes dish soap, hand soap, and stain remover. The dispenser is still plastic but designed to work for years. Puracy’s refills, however, get packaged in aluminum cans, similar to soft drinks and beer. The refill process takes seconds. The consumer pops open the can, snaps it to the bottom of the pump, and twists the second piece of the dispenser to the pump. The refills are easy to store, stock, and use and added advantage is that the packaging material i.e. aluminum is recyclable.

Refill Revolution

New-age materials are popularly being used to make refills, reducing the use of plastics even further. Bamboo pulp refills are popular as Bamboo is a low maintenance plant that grows quickly. The paper product created from its fibers are easily biodegradable. Hybrid packaging solutions using bamboo as part of the material are being used by some of the world’s most recallable brands, such as The Body Shop as well as startups that are incorporating sustainability as part of their brand story. Product design is also experiencing the impact of the ‘refill revolution’.

Design for refill packaging itself can be quite minimalistic as it is not the packaging that is required to draw the consumer in. For example, French brand L'Occitane which has a very distinct signature design aesthetic, hosts product refills for shower gels, shampoos, conditioners, and liquid soaps in simple packaging. While the refillable bottle – which is the product that is primarily responsible for customer interaction – continues to host their signature style, the refills are visibly different in their look and feel.



9.1: KanKan's refillable aluminum cans redefine eco-friendly body care packaging with a sleek, sustainable touch. | 9.2: Morrama designed 100% plastic-free, compostable, liquid cosmetics refill from bamboo and plant starch | 9.3: Berlin Packaging has created chic and yet eco-friendly bottles designed for easy recycling and refillability for the luxury fragrance market | 9.4: Caudalie innovates with refillable jars for skincare, combining elegance with sustainability.

The Bottom Line

Packaging for refills – materials and design – doesn’t have a defined playbook and so, new precedents are fast becoming the norm. Within these changes are tremendous opportunities for innovation in a way that complements sustainability objectives.

Experts Speak

Packforum 2024

Packforum 2024 was held on June 5, 2024 at New Jersey, USA. We brought together industry leaders to engage in topics ranging from sustainability to new-age packaging materials. This evoked dialogue and idea sharing, resulting in new insights. Packforum's mission is to turn challenges into opportunities by leveraging collective strengths.

Companies at Packforum included leadership from Danone, Mondelez, Amazon, Unilever, Bayer, Pepsico, and more. The combined revenue of the participating organizations was upwards of \$500 billion, with a total packaging spend of \$30 billion.

Packforum Philosophy

Understand



*What are
the future
forces?*

Unlock



*Through
which avenues
can we create
new value?*

Co-create



*Are there
ways for us to
collaborate?*



We need to stay on top of the changing materials and recycling landscape, making steady progress along the way.

Kory Nook

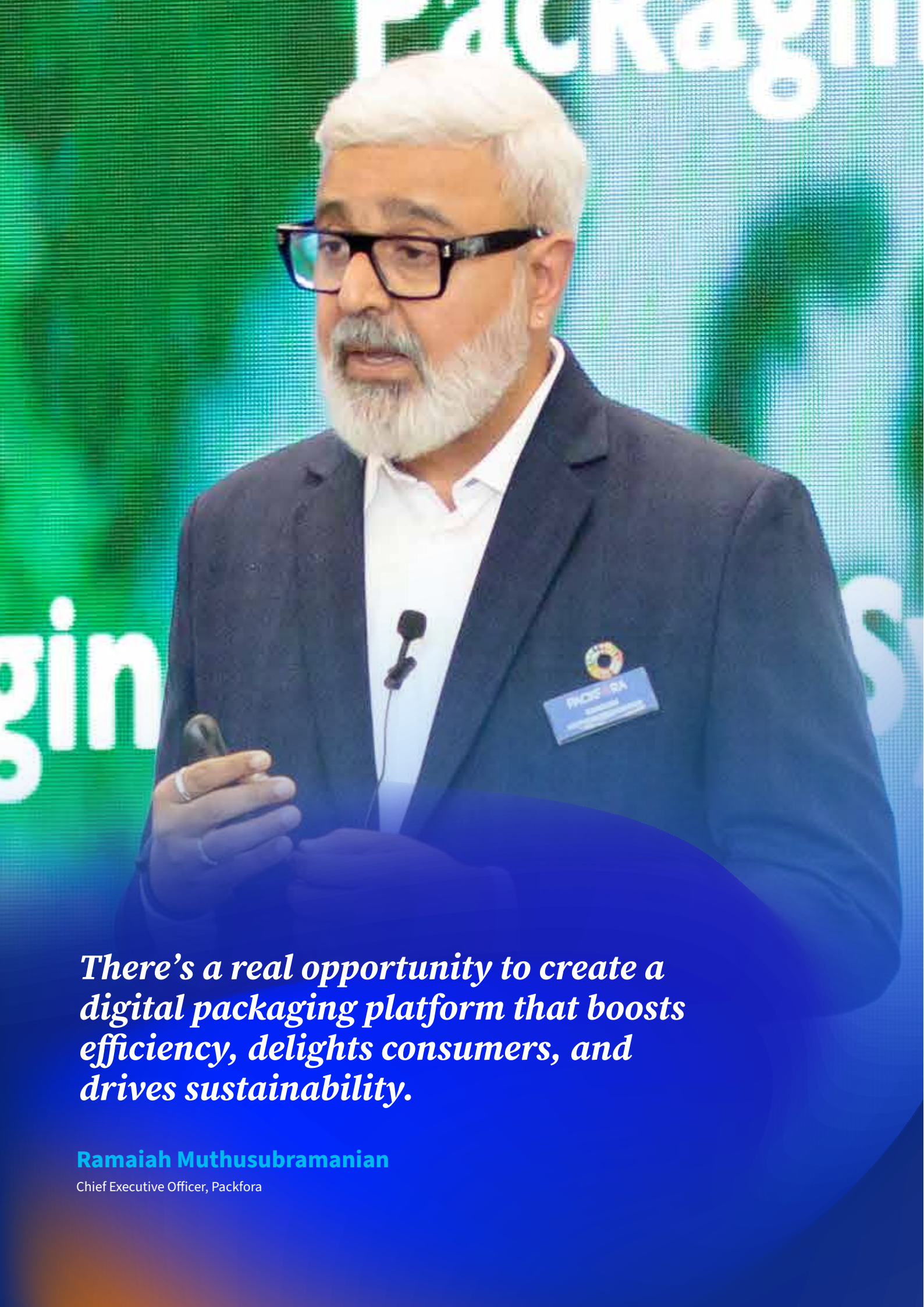
Vice-President (R&I)
Danone

New Materials & Sustainability

Kory Nook, Vice-President (R&I) at Danone spoke extensively about new materials and sustainability in packaging. The main focus of his discussion was explaining that our planet is burning, and as businesses, we must respond to this, by changing how we operate.

Important takeaways from his discussion included:

- **Change needs to happen NOW:** He emphasized an urgent need for innovation across the value chain to address environmental concerns. Current practices are unsustainable and could harm the businesses if not changed.
- **Pressure through regulations:** Emerging regulations in the US and EU are driving the need for sustainable packaging solutions. Companies must comply with stringent laws and meet investor and consumer expectations around environmental, social, and governance (ESG) metrics.
- **Investors care about decarbonization:** Investors are increasingly prioritizing decarbonization and are critical of greenwashing. To attract capital, it's imperative that companies set clear CO2 reduction targets and integrate them into their business strategies.
- **The consumer cares and the consumer knows:** Consumers are becoming more sophisticated in their understanding of sustainability. At the same time, the consumer shows an unwillingness to pay more for eco-friendly products. Companies need to innovate to match consumer sentiments without significantly increasing costs.
- **Create a competitive edge:** Companies that lead in sustainable practices can gain an advantage because there is significant growth potential in sustainable products, with consumers increasingly driven by environmental and social concerns. This can be achieved through multiple steps, including evaluating every step of the value chain from sourcing to end-of-life and valorizing through the brands and tie to product innovation.



There's a real opportunity to create a digital packaging platform that boosts efficiency, delights consumers, and drives sustainability.

Ramaiah Muthusubramanian

Chief Executive Officer, Packfora

Connected Commerce & Smart Packaging

Ramaiah Muthusubramanian, Chief Executive Officer at Packfora explained how there is an opportunity to creating common packaging digital platform. This, he discussed, will meet diverse requirements from consumer delight to supply chain efficiencies to packaging disposal.

Important takeaways from his discussion included:

- **Connected Commerce:** The convergence of online and offline retail experiences is transforming consumer interactions. Technologies like IoT, AI, and data analytics are central to this evolution as they are able to facilitate personalized and efficient shopping experiences.
- **Current Trends:** The global use of connected devices is growing exponentially, with an estimated 25 billion gadgets in use worldwide. E-commerce is expected to follow a similar trend wherein estimates say it will grow by 75%. As could be imagined, Gen Z consumers are particularly influenced by digital discovery of products, often turning to social media and search engines rather than browsing physical store shelves.
- **Technological Innovations in Brand Experience:** Companies are focusing on building brand experiences that bridge the virtual and real worlds, enhancing consumer connections through personalization. The use of blockchain for authenticating luxury goods and ensuring a trustworthy supply chain is highlighted. A terrific example of this the Aura software being used by Prada.
Each product is assigned a unique digital certificate that is stored on the blockchain. When a customer purchases a Prada item, they can access the product's digital certificate by scanning a QR code or NFC tag on the product. This certificate provides information regarding the entire product's journey from sourcing of materials to point of sale. In an industry where counterfeits present a huge problem, this is an extremely disruptive solution.
- **Sustainable and Smart Packaging:** There is a growing emphasis on the role of packaging in the consumer experience. We're seeing innovations like QR-enabled disposal instructions and technologies like QR 2.0, NFC, RFID, and printed electronics being explored. Future packaging design will continue to prioritize sustainability, including better recycling practices and managing global packaging taxes.

We are likely to soon find ourselves in a space where digital and physical worlds entirely converge, creating a seamless, technology-driven consumer experience.

The integration of advanced technologies into packaging will play a crucial role in maintaining the "first moment of truth" with consumers in a digital-first world. This is why embracing digital transformation in commerce and packaging to meet evolving consumer expectations and drive innovation is not only important but also essential.



Zero-based design is about making conscious choices that bring together science, branding, and consumer insights to create the best packaging.

Abhay Bhagwat

Managing Consultant,
Packaging for the Future

Zero-Based Design

The value of 'zero-based design' (ZBD) was covered extensively by Abhay Bhagwat, Managing Consultant at Packaging for the Future. He explained that efficient packaging solutions used a 'first principles' approach which focused on minimizing material usage, reducing costs, and optimizing the entire supply chain. ZBD is a stage-gate process that improves the phase of generation of design options.

Important takeaways from his discussion included:

➤ **Main concepts of ZBD:** Minimum Viable Pack (MVP): The simplest possible packaging that meets essential requirements. This includes aesthetics, material efficiency, cost-effectiveness, and supply chain efficiency. It involves firstly looking the cost equation which is \$ per ton of finished product, converting into an equation so we can list all the key design variables that are impacting the cost. The MVP is important as it allows you to innovate in two ways:

1. By adding new relevant design features.
2. By retaining iconic features but subtract non value added elements, which ultimately gives "Value".

Design Variables: Bottle shape, size, and material – concentrating on these is the path towards achieving maximum packaging efficiency.

➤ **Demystifying the ZBD process:** The ZBD process involves defining cost equations, modeling bottle geometry, and optimizing design based on shape factors and material properties. This results in a "Minimum Viable Bottle" (MVB), which can be further refined or simplified to meet specific brand needs.

➤ **Key equations to be considered:**

- Bottle cost equation: This optimises the bottle cost considering the key variables that can influence and eventually improve the grams (weight) of the bottle per ml.
- Plastic Equation: Then gram/ml is broken into relevant variables to optimise the volume of plastic per bottle.

➤ **Bottle optimization:** The overarching focus for ZBD is to create a balance between what a product looks like and what it is made from. Factors such as bottle shape may not seem important but play a critical role in driving efficiency. ZBD principles highlight how even the smallest improvements in bottle shape can significantly reduce the bottle's overall weight, leading to lower costs.

➤ **The value of ZBD:** At the core of it, ZBD is not meant to replace traditional design methodologies, but to enhance them. In doing, the process strengthens brand identity, optimizes performance across the supply chain, reduces environmental impact, and delivers cost savings with low capital expenditure.

Simply put, ZBD is in no way a one-size fits all strategy, instead it has to be technically optimized across all touch points in the existing value chain, as is relevant to the product and the brand.



Inclusive thinking starts with the brief—no pack is perfect; design must be driven by real consumer needs.

Brett Domoy

Head of Packaging & Process Engineering
Skin Cleansing, Unilever

Inclusive Innovation

Brett Domoy, Unilever – Head of Packaging and Process Engineering spoke extensively about product design's role in improving inclusivity of packaging. He explained that the global numbers of people classified as differently abled is huge and every organization will need to adopt inclusive design, either today or in the near future.

Important takeaways from his discussion included:

➤ **Understanding impairments:** Developmental impairments are essentially any conditions of the mind or body that makes it more difficult for said person to interact with the world around them. Some impairments are not apparent or visible to the untrained eye.

➤ **Relevance to today's demographic:** The presence of those needing inclusive packaging is significant; for example, data across sources such as UNESCO, WHO and the National Eye Institute and more, estimate the following about the global population:

700 Million people are dyslexic

2.2 Billion are visually impaired

773 Million are illiterate

300 Million people are color-blind

1.5 Billion have low reading skills

1 Billion people have some form of physical disability

Adding to these figures is the elderly population which battles with disabilities that accompany aging.

➤ **Design for impairments vs design for inclusivity:** These are two very different things – a fitting example would be a time piece that is created with numbers in Braille. While this responds to blindness per se, it is not particularly inclusive as fewer than 1% of blind people use Braille. In contrast, a time piece that was designed using a universal tactile system made life easier for anyone who is unable to see their watch.

➤ **Design factors:** It is critical to look at the overall usability, including aspects like on-pack hierarchy, font size, colors, imagery, iconography, typeface, tactile indicators, and packaging shape. Specific design considerations can be achieved by taking into account the following considerations:

1. Color as a Differentiator: This helps users navigate product ranges quickly and can influence perception. At the same time, creators must bear in mind that color usage may send incorrect signals, if not used carefully.

2. Imagery and Iconography: Strategic usage of visuals conveys information, reduces text clutter, and creates connections with consumers across different languages and cultures.

3. Typeface and Language: Paying attention to font choices advocates for readability with simple, clear instructions and accessible color contrasts.

4. Packaging Shape: Shape is important as it includes considerations like waisted design, hand/finger indents, and pump packs to enhance usability for a broader audience. For example, some packages are easier to hold and/or squeeze.

The main thing to remember when trying to create inclusive packaging is that there is no perfect pack. Simply striving for inclusive design will lead to better, more universally usable products.

Wrapping Up

The packaging industry is evolving rapidly, influenced by digital advancements, consumer preferences, and sustainability regulations. Our immediate deliverable is establishing a balance between sustainability and customer engagement, strategic packaging design, adapting to rapid changes, purpose-driven innovation, and creating value through social media engagement.

As we look towards what's next, we must try and create packaging solutions that are future-fit, by way of not only technology incorporation but also the dynamic nature of customer sentiments.

Solutions must be easily adaptable and scalable. Simply, preparing for future forces is possible only through being able to identify their potential to create new value.

Testimonials



Kory Nook
Vice-President (R&I)
Danone

Having a platform and a forum like this re-inspires us to connect again and figure out how we can make a bigger transformational leap with some of our packaging portfolios.



Kelly Dawson
Strategy Partner
& Design Partner,
Studio Every

I was blown away actually by the engagement. The intimacy and the size enabled people to raise their hand, give their views. I love that, the intimacy.



Brett Domoy
Head of Packaging
& Process Engineering
Skin Cleansing,
Unilever

I thought it was absolutely incredible. Packforum was able to assemble a group of industry professionals, like minded individuals who all have the ability to impact the industry.



Jon Genord
Director – Packaging,
Danone

Packfora brings us a lot of value. It's exciting to be here and have a dedicated audience, not only with the Packfora team, but industry leaders across their network.





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